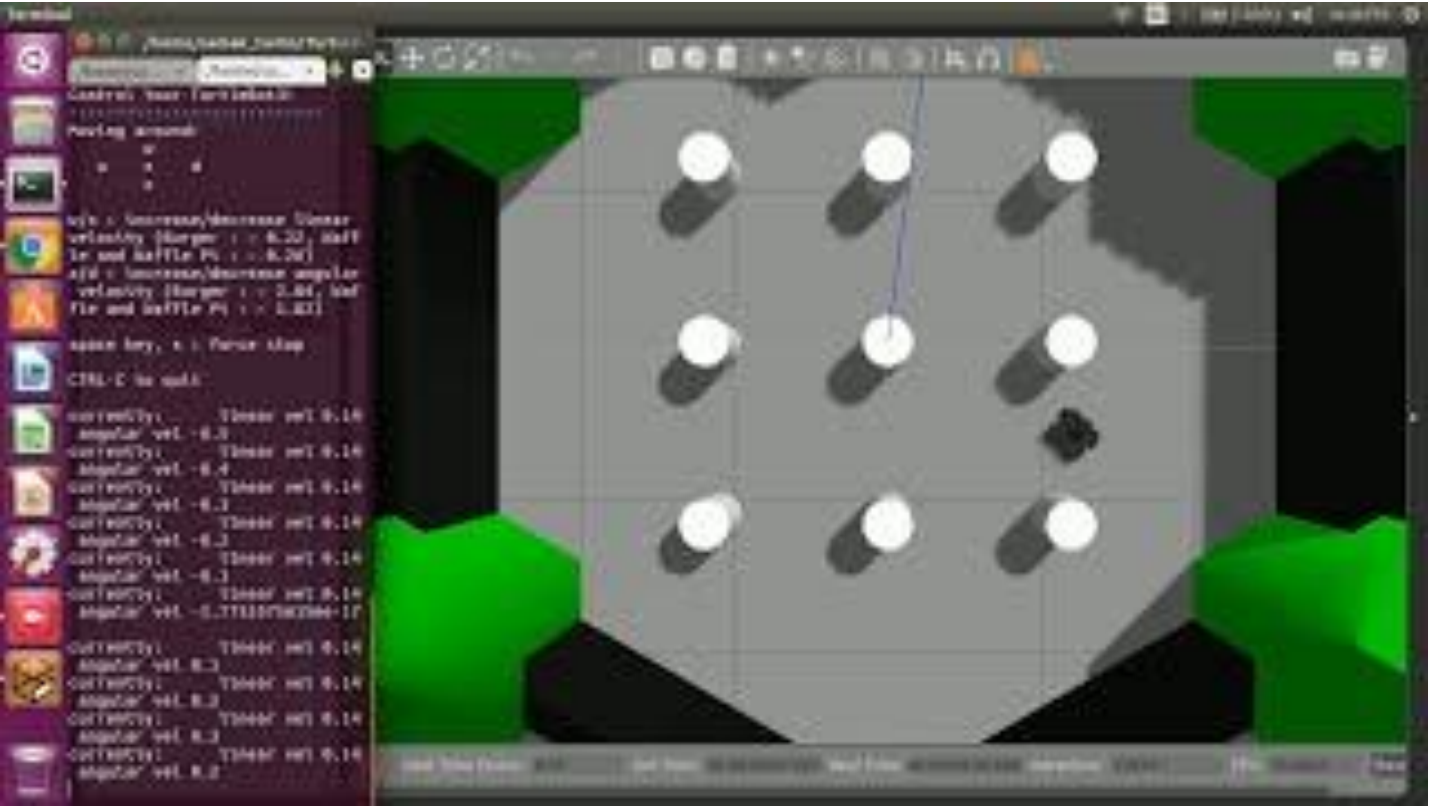
Mobile Robotics

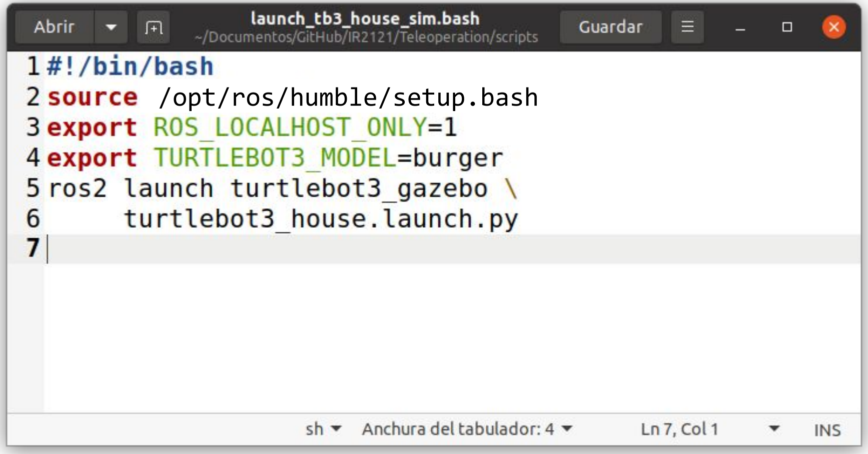


Bachelor's Degree in Robotic Intelligence

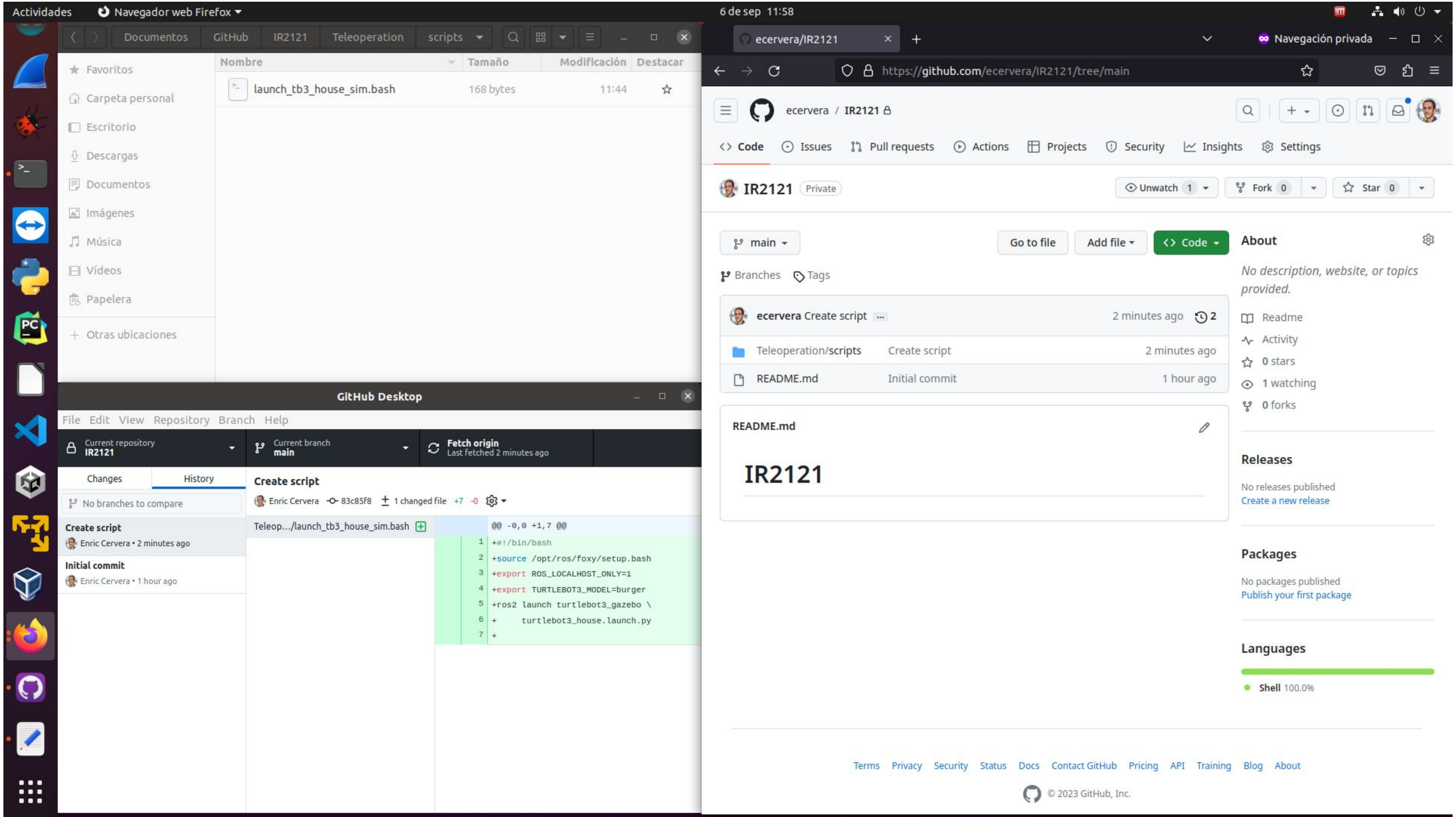
Teleoperation

Teleoperation of TurtleBot 3 in ROS Humble and Gazebo

1. Create a folder named "`Teleoperation`" in your repository.
2. Create a folder named "`scripts`" inside the "Teleoperation" folder.
3. Create a BASH script named "`launch_tb3_house_sim.bash`" with the following lines
4. Change the script properties (**<u>permission to execute</u>**).
5. Commit and push the repository.
6. Test the script.



Teleoperation of TurtleBot 3 in ROS Humble and Gazebo

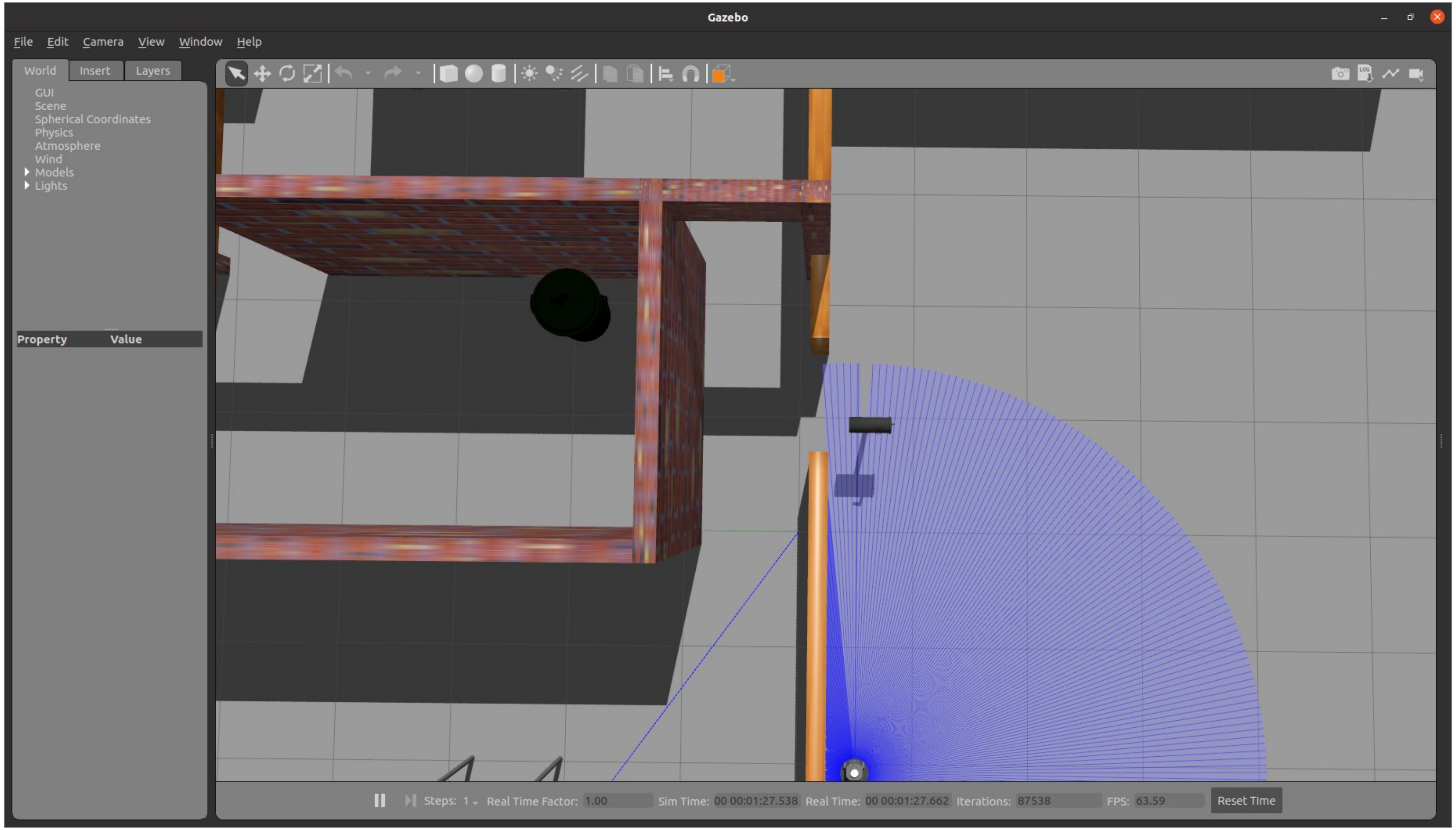


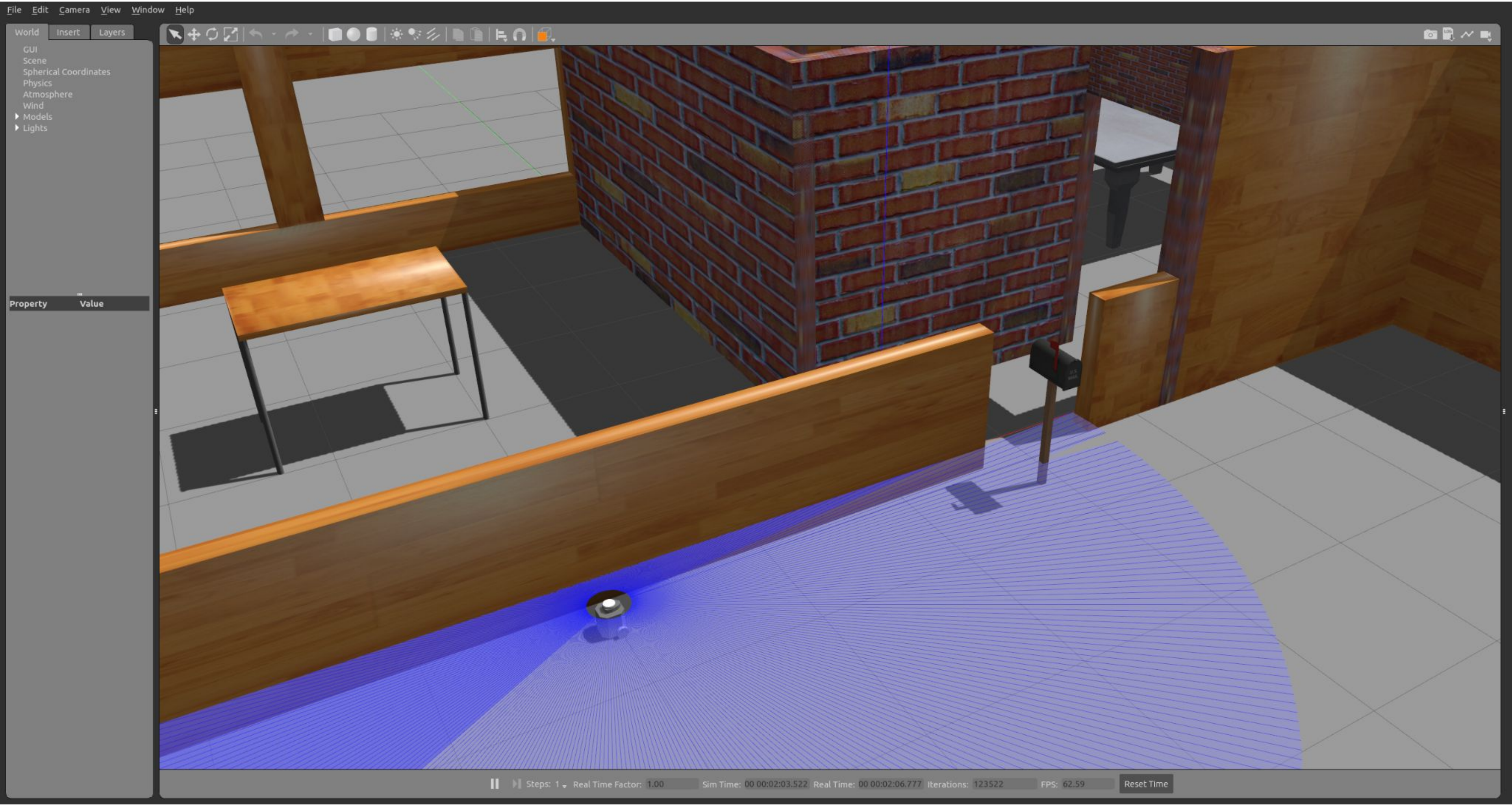
Teleoperation of TurtleBot 3 in ROS Humble and Gazebo

usuario@LABROB06: ~/Documentos/GitHub/IR2121/Teleoperation/scripts

```
usuario@LABROB06:~/Documentos/GitHub/IR2121/Teleoperation/scripts$ ./launch_tb3_house_sim.bash
[INFO] [launch]: All log files can be found below /home/usuario/.ros/log/2023-09-06-11-52-00-845637-LABROB06-9392
[INFO] [launch]: Default logging verbosity is set to INFO
urdf_file_name : turtlebot3_burger.urdf
[INFO] [gzserver-1]: process started with pid [9394]
[INFO] [gzclient   -2]: process started with pid [9397]
[INFO] [robot_state_publisher-3]: process started with pid [9400]
[robot_state_publisher-3] [WARN] [1693993921.175503986] [robot_state_publisher]: No robot_description parameter, but
 command-line argument available.  Assuming argument is name of URDF file.  This backwards compatibility fallback wi
ll be removed in the future.
[robot_state_publisher-3] Parsing robot urdf xml string.
[robot_state_publisher-3] Link base_link had 5 children
[robot_state_publisher-3] Link caster_back_link had 0 children
[robot_state_publisher-3] Link imu_link had 0 children
[robot_state_publisher-3] Link base_scan had 0 children
[robot_state_publisher-3] Link wheel_left_link had 0 children
[robot_state_publisher-3] Link wheel_right_link had 0 children
[robot_state_publisher-3] [INFO] [1693993921.176444524] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-3] [INFO] [1693993921.176455327] [robot_state_publisher]: got segment base_link
[robot_state_publisher-3] [INFO] [1693993921.176458825] [robot_state_publisher]: got segment base_scan
[robot_state_publisher-3] [INFO] [1693993921.176461749] [robot_state_publisher]: got segment caster_back_link
[robot_state_publisher-3] [INFO] [1693993921.176464663] [robot_state_publisher]: got segment imu_link
[robot_state_publisher-3] [INFO] [1693993921.176467477] [robot_state_publisher]: got segment wheel_left_link
[robot_state_publisher-3] [INFO] [1693993921.176470331] [robot_state_publisher]: got segment wheel_right_link
[gzserver-1] Warning [parser.cc:833] XML Attribute[version] in element[sdf] not defined in SDF, ignoring.
[gzserver-1] [INFO] [1693993922.638846323] [turtlebot3_imu]: <initial_orientation_as_reference> is unset, using defa
ult value of false to comply with REP 145 (world as orientation reference)
[gzserver-1] [INFO] [1693993922.664157108] [turtlebot3_diff_drive]: Wheel pair 1 separation set to [0.160000m]
[gzserver-1] [INFO] [1693993922.664202167] [turtlebot3_diff_drive]: Wheel pair 1 diameter set to [0.066000m]
[gzserver-1] [INFO] [1693993922.664953963] [turtlebot3_diff_drive]: Subscribed to [/cmd_vel]
[gzserver-1] [INFO] [1693993922.666052752] [turtlebot3_diff_drive]: Advertise odometry on [/odom]
[gzserver-1] [INFO] [1693993922.666604940] [turtlebot3_diff_drive]: Publishing odom transforms between [odom] and [b
ase_footprint]
[gzserver-1] [INFO] [1693993922.673107578] [turtlebot3_joint_state]: Going to publish joint [wheel_left_joint]
[gzserver-1] [INFO] [1693993922.673139798] [turtlebot3_joint_state]: Going to publish joint [wheel_right_joint]
```

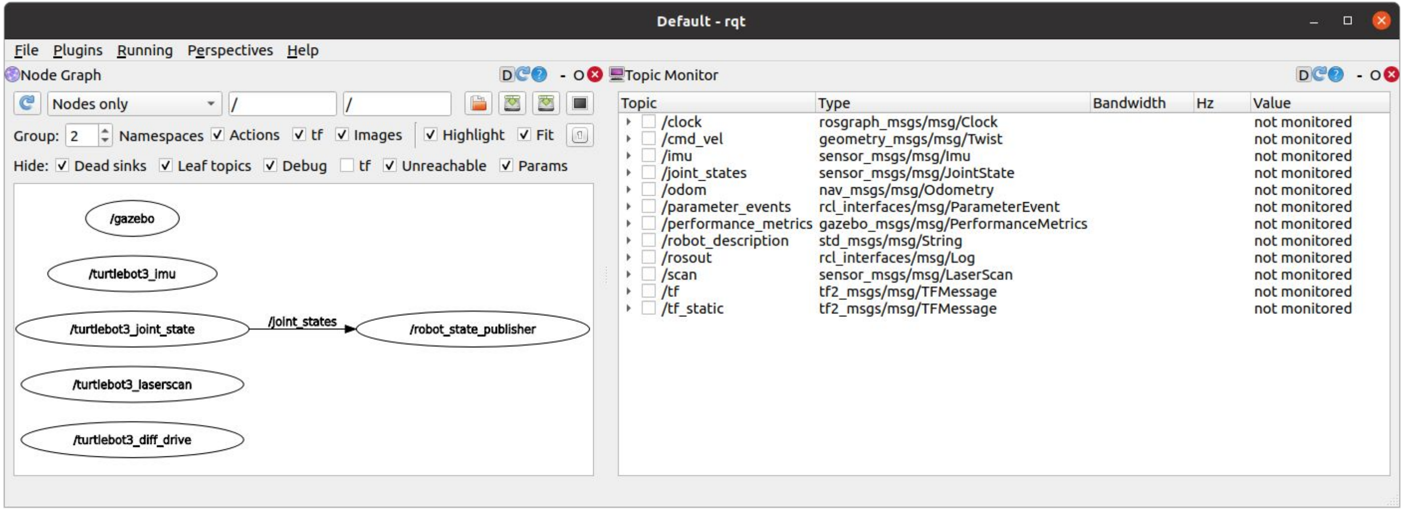
Teleoperation of TurtleBot 3 in ROS Humble and Gazebo



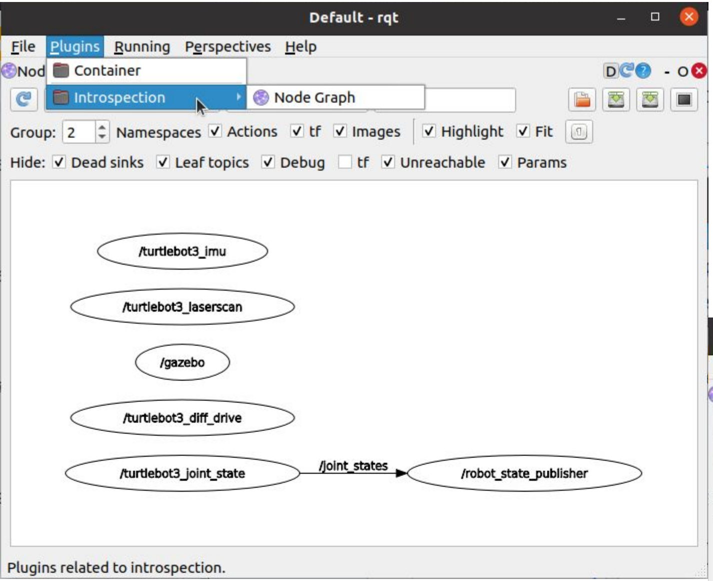

File
Edit
Camera
View
Window
Help
World
Insert
Layers
GUI
Scene
Spherical Coordinates
Physics
Atmosphere
Wind
Models
Lights
Property
Value
Steps: 1
Real Time Factor: 1.00
Sim Time: 00 00:02:03.522
Real Time: 00 00:02:06.777
Iterations: 123522
FPS: 62.59
Reset Time

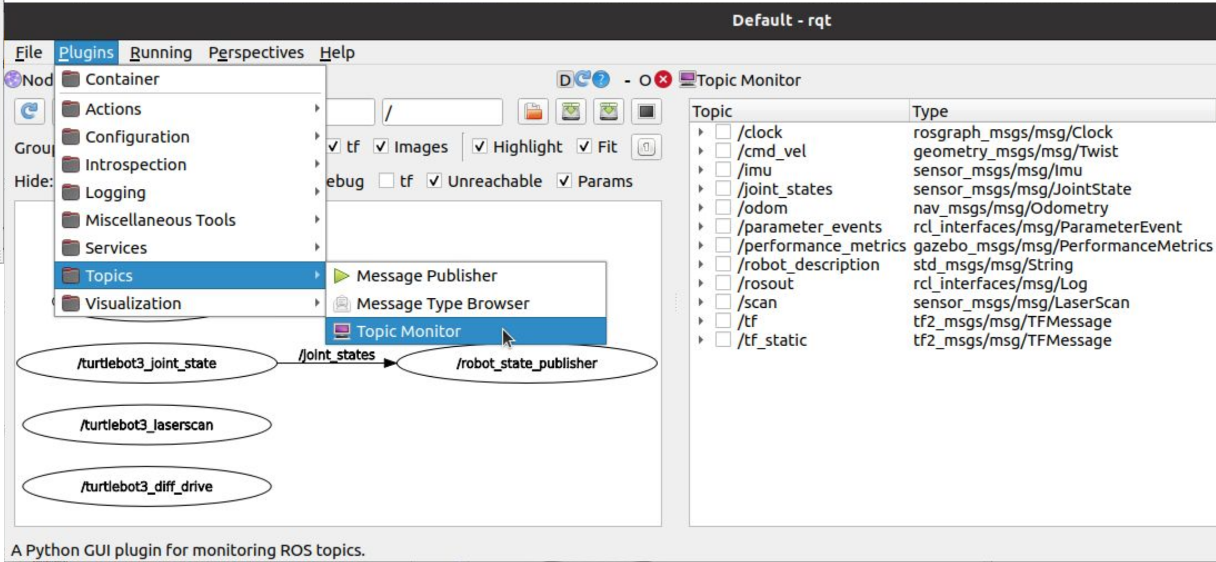
Visualization

1. Create another BASH script named “`launch_rqt.bash`” in the `scripts` folder.
2. Change the script properties (permission to execute).
3. Commit and push the repository.
4. Test the script.



Visualization

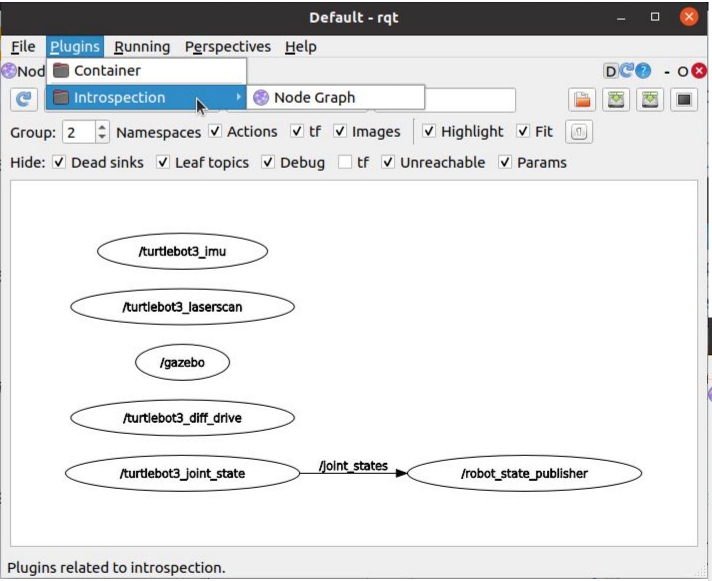


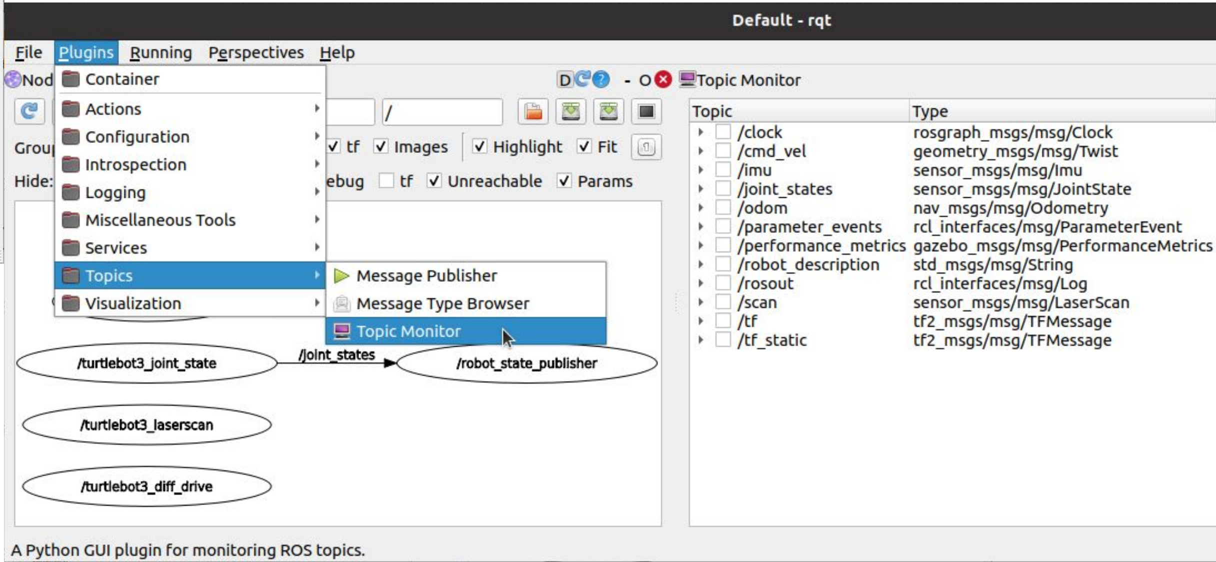


Missing rqt plugins

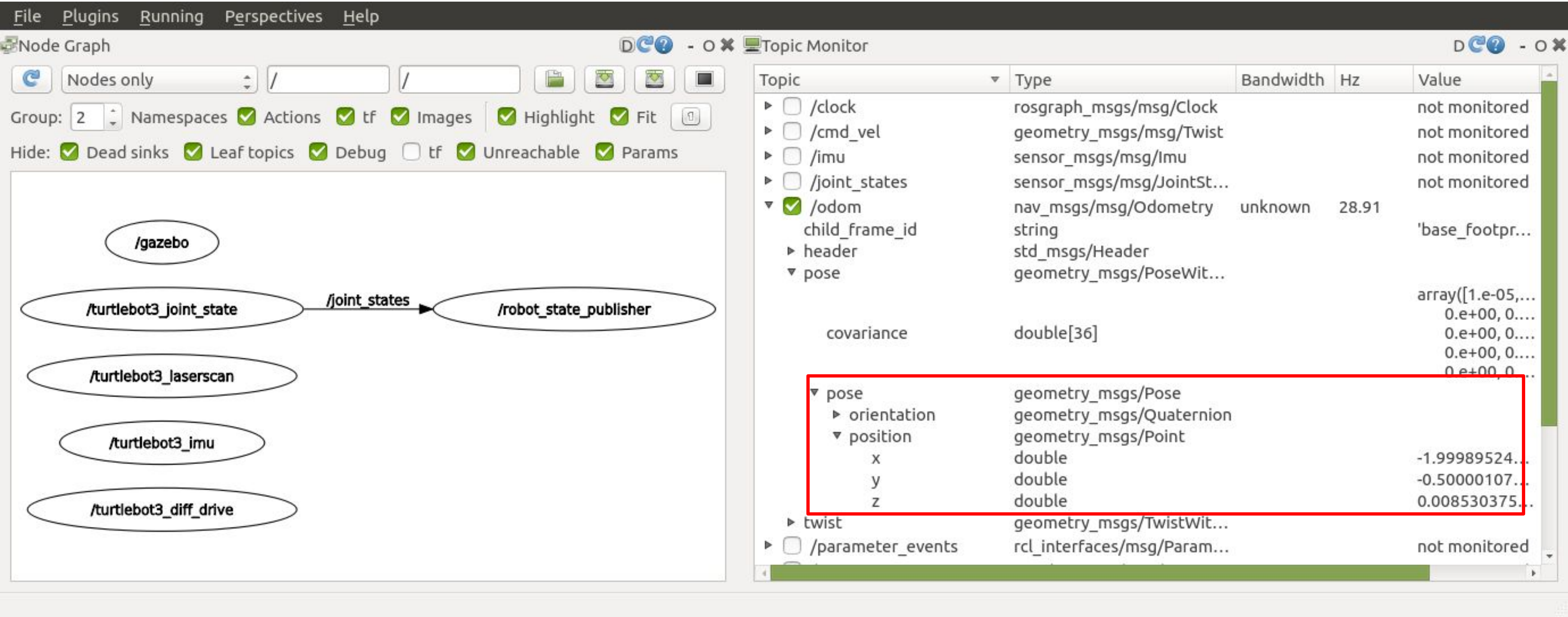
```
sudo apt update

sudo apt install \
ros-humble-rqt-common-plugins
```

Launch `rqt` with `--force-discover` option.



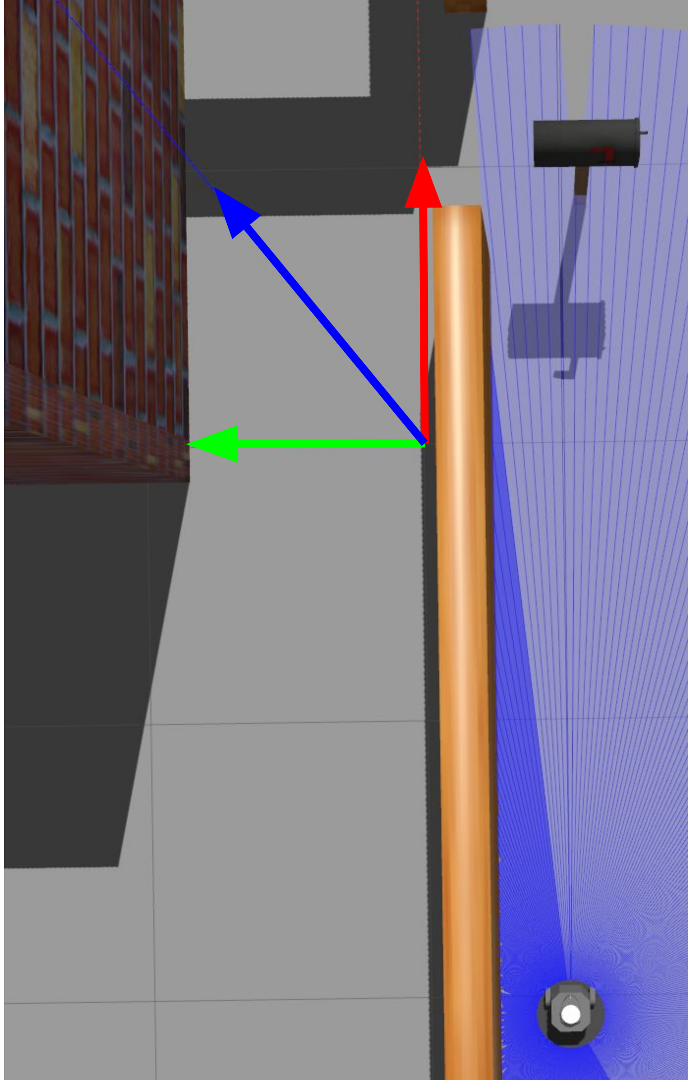


Odometry topic



Origin of coordinates

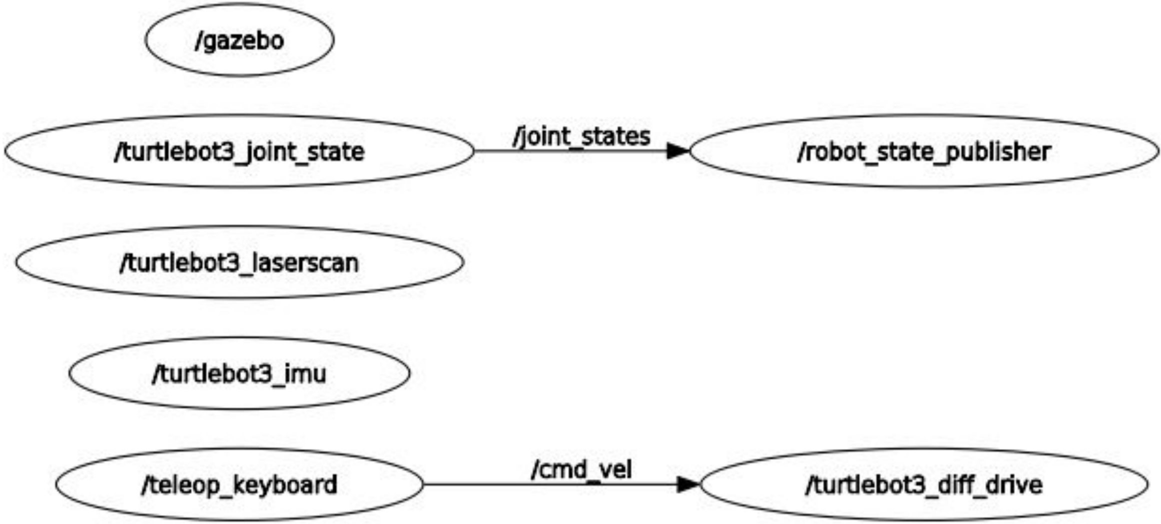
```
Pose of robot

    x   -1.9998

    y   -0.5000

    z    0.0085
```



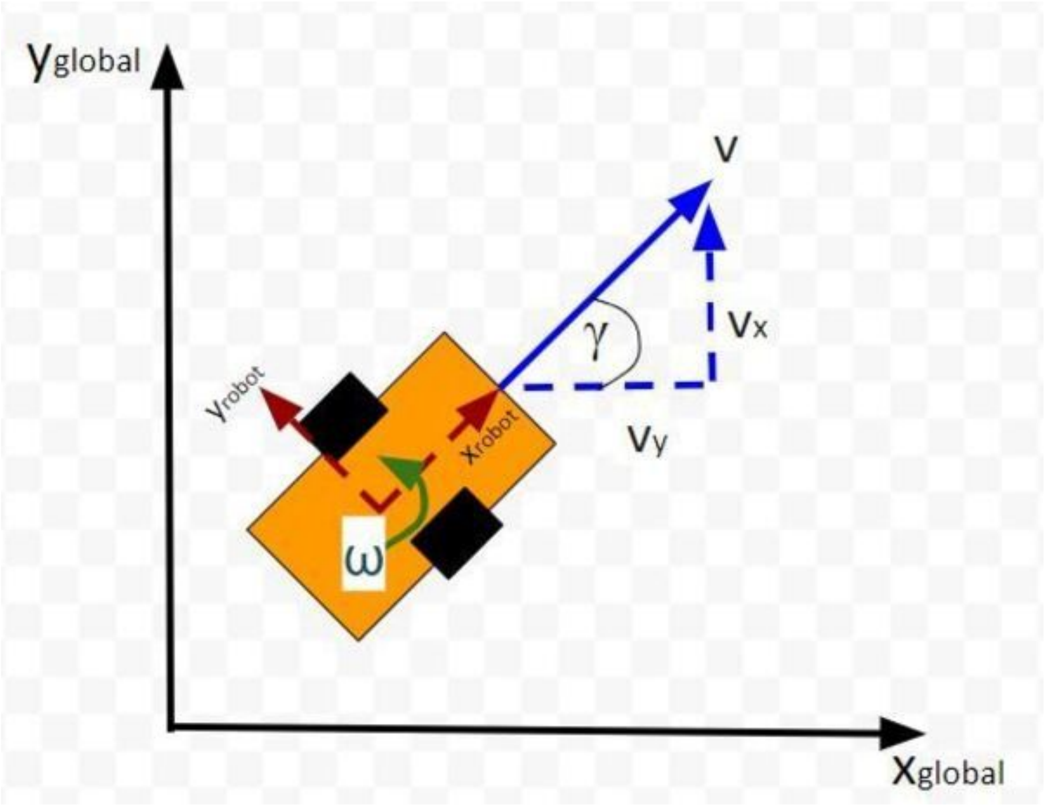
Teleoperation

1. Create another BASH script named "`launch_tb3_teleop.bash`" in the scripts folder.
2. Change the script properties (permission to execute).
3. Commit and push the repository.
4. Test the script.

```
Control Your TurtleBot3!
---------------------------
Moving around:
        w
   a    s    d
        x

w/x : increase/decrease linear velocity
(Burger : ~ 0.22, Waffle Pi : ~ 0.26)
a/d : increase/decrease angular velocity
(Burger : ~ 2.84, Waffle Pi : ~ 1.82)

space key, s : force stop

CTRL-C to quit
```



Linear and angular velocities



Task

Teleoperate the robot

Visit all the rooms

Finish at the same initial location

Record topics with `ros2 bag record`

Plot trajectory in RViz

ROS bags

Record

```
source /opt/ros/humble/setup.bash
export ROS_LOCALHOST_ONLY=1
ros2 bag record /clock /odom /tf /tf_static
```

Play

```
ros2 bag play rosbag2_2025_...
```

[Recording and playing back data — ROS 2 Documentation]()

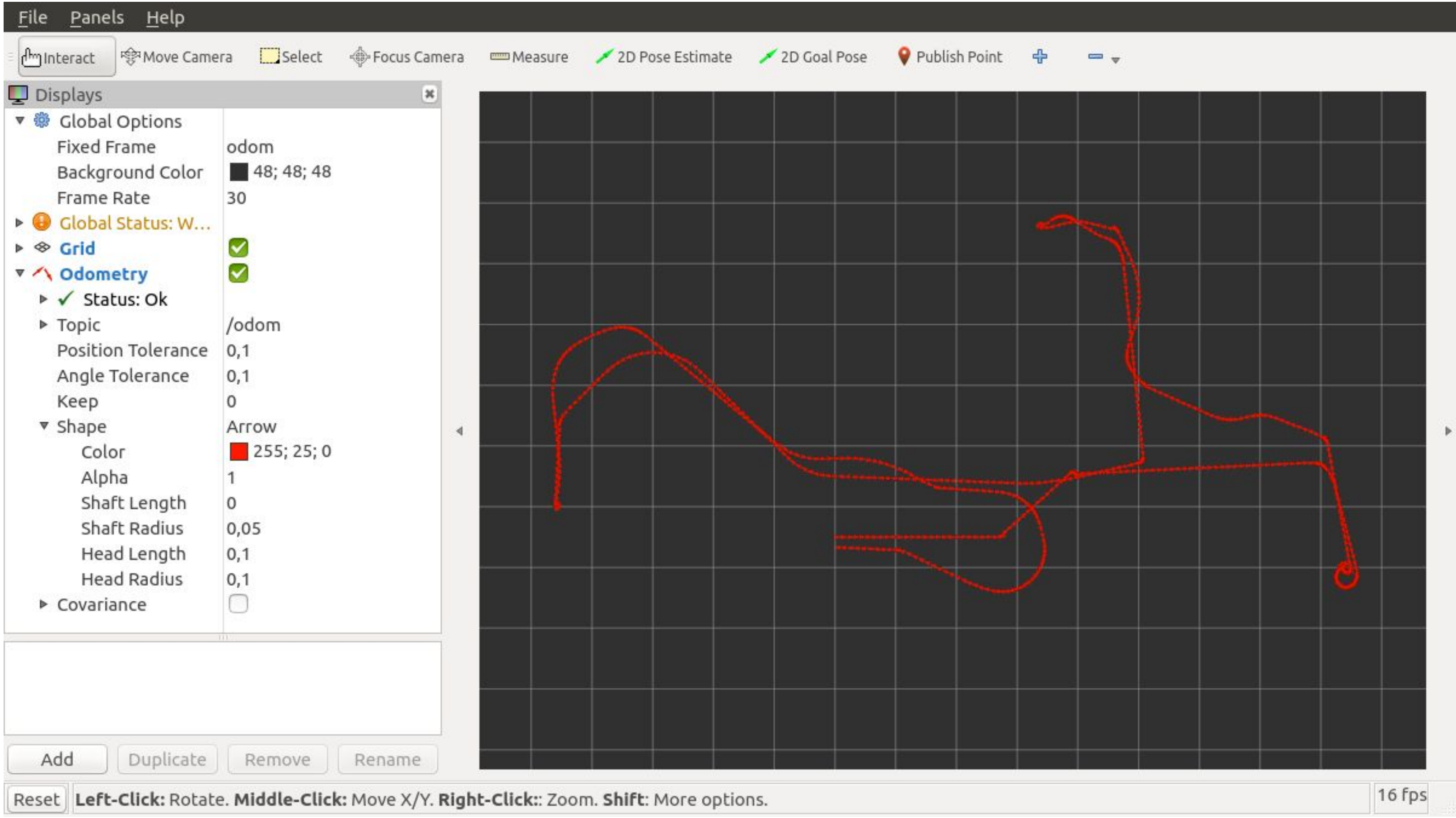
Analyze bag files

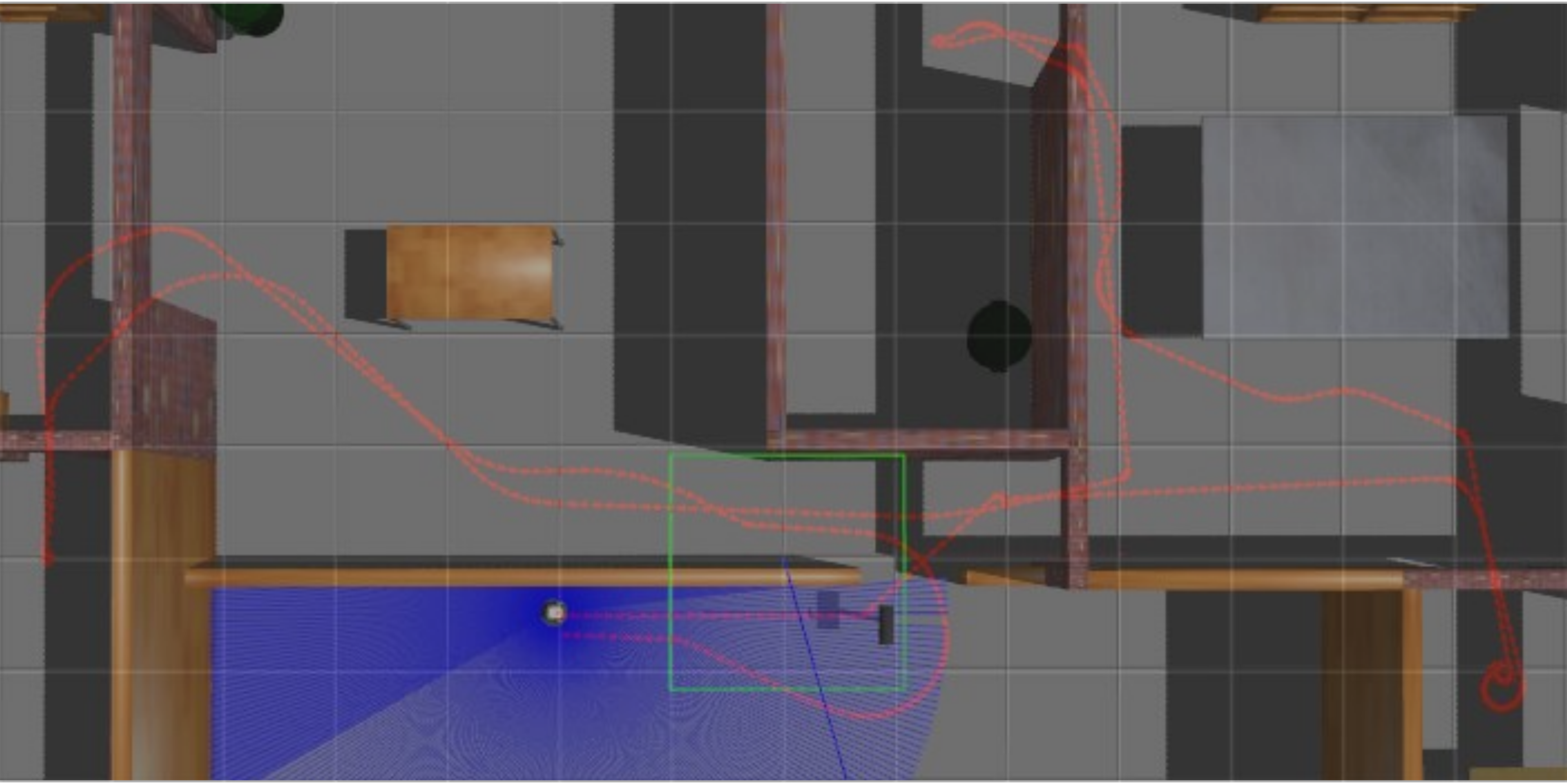
```
ros2 bag info <bagfile>
```

- Bag size
- Duration
- Number of messages
- Topic information:
  - topic
  - type
  - count

RViz

```
rviz2 -d config.rviz
```





Assignment

Submit to the Aula Virtual the following files:

- ZIP archive of bag folder
- script for launching ros2 bag play
- script for launching rviz
- rviz config file
- picture of the robot’s trajectory on top of a simulator image

Recording and Playing

Recording	**Playing**
Gazebo	~~Gazebo~~
Teleop	~~Teleop~~
ros2 bag **record** …	ros2 bag **play** …
rviz2	rviz2